Math 55 worksheet Monday 02/23/2026

1. Prove that a positive integer n is a perfect square if and only if every exponent in its prime factorization is even.

2. Let p be a prime number, and let $\nu_p(n)$ denote the exponent of p in the prime factorization of n. If a and b are any two positive integers, prove that

$$\nu_p(a+b) \geq \min(\nu_p(a), \nu_p(b)).$$

Furthermore, if $\nu_p(a) \neq \nu_p(b)$, prove that

$$\nu_p(a+b) = \min(\nu_p(a), \nu_p(b)).$$

3. Let a and b be positive integers with prime factorizations $a = p_1^{e_1} \cdots p_k^{e_k}$ and $b = p_1^{f_1} \cdots p_k^{f_k}$. Show that (a)

$$\gcd(a,b) = p_1^{\min(e_1,f_1)} \cdots p_k^{\min(e_k,f_k)},$$

(b)

$$\operatorname{lcm}(a,b) = p_1^{\max(e_1,f_1)} \cdots p_k^{\max(e_k,f_k)},$$

and deduce (c):

$$ab = \gcd(a,b)\operatorname{lcm}(a,b)$$